

Gold Price Prediction using Random Forest Algorithm

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Abstract—Gold price is one of the most important and influential factors for a nation's economy or its stock market. This research work has carried out an extensive study to interpret the correlation between the gold price and a bunch of other parameters. The parameters that we have taken into consideration are Date, SPX, USO, SLV, EUR/USD, and GLD. Out of all of them, our target variable is the GLD which is the Gold price for the given date which depends on all other factors. SPX is short for the S&P 500, which is one of the most well-known stock market indexes. About 500 of the biggest corporations listed on the American stock market are included in it. USO represents the United States Oil price for the given date. SLV is the current price of silver on any given day. The abbreviation for the cross between the currencies of the European Union (EU) and the United States is the currency pair EUR/USD (USD). The currency pair displays the amount of US dollars (the quote currency) required to buy one euro. We have trained a Random Forest model to predict the Gold prices based on the value of SPX, USO, SLV, and EUR/USD.

Keywords- Gold Price, Machine Learning, Artificial Intelligence, Random Forest Algorithm, Algorithm

I. INTRODUCTION

The ability to predict the future trend of gold prices is crucial in the financial sector. Profits can be generated by even little improvements in predicting performance. Finance is a necessary aspect of life, and accurate financial forecasting will also help the economy grow.

These days, economists and other researchers use some mathematical techniques to examine financial data to make precise predictions that may or may not be useful. Almost all established cultures have utilized gold as a symbol of power, beauty, purity, and success throughout history. As compared to gold, no other substance is that prominent and expensive. Because of its rarity and high value, it is being used as a form of currency. As gold reserves demonstrate the country's economic strength, it has inspired nations to accumulate gold to strengthen their finances. A lot of factors influence the price of gold, resulting in price volatility. Inflation, supply and demand, and political difficulties are just a few of the factors at work. When inflation rises, the gold price rises correspondingly, also when it is scarce, its price rises.

The graph depicts the fluctuation of the gold price over 41 years, from March 1978 to November 2018. We can see the volatility of the gold price. This fluctuation may have been the result of a lot of factors and economic conditions like recession, wars, etc.

Many researchers have employed a variety of approaches and considered different factors to get the best possible and accurate results. A few of the variables considered to improve the prediction are the US dollar index (USDIX), crude oil price (COP), Dow Jones Industrial Average (DJIA), US CPI (USCPI), US ten-year bond futures prices (US10BFP), Hang Seng Index (HIS), Standard & Poor's 500 Index (S&P500), etc. However, gathering data becomes more challenging the more variables that need to be taken into account.

The link between the gold value and various other economic indicators has been the subject of several studies. Understanding the relationship between these aspects can help investors make better decisions. The Random Forest Machine Learning algorithm, which is frequently utilized in regression and classification issues, is applied in our study. Nonetheless, it's critical to recognize the drawbacks of utilizing the Facebook Prophet algorithm to predict gold prices[8]. The Random Forest Algorithm could be able to handle various data sets which can be with both continuous as well as categorical variables, such as the regressions and the classifications. Gold's price is also linked to the SP200, USO (oil price), SLV (silver price), and EUR/USD.

II. LITERATURE SURVEY

Makala and Z Li [1] have applied SVM and ARIMA models respectively as a means to predict the gold price. Their study used gold price data over 35 years (1979 – 2014) for training the models and 6 years' worth of data (2014 – 19) from the archives of the World Gold Council (WGC) to test their model. ARIMA or Autoregressive Integrated Moving Average model is a type of univariate time-series-based regression model that modifies a changing variable based on its lagged/past data. ARIMA is a popular technique for gold and stock price prediction as it performs well on a series of gradual, mostly stationary data having > 100 observations [2] [3]. On data where there are seasonal changes or sharp rises

and falls within specific periods, this model doesn't perform as well as its other counterparts. The researchers used a (2,1,2) model and compared it with SVM (linear, poly, and rbf). Their results showed that SVM (poly) gave the best results, an R2 score of 99% as well as an RMSE of just 0.0275 as compared to the ARIMA model which gave an R2 score of 86% and RMSE of 36.1795. However, the poly kernel used only showed the best results with a degree = 1.9, which may have been overfitting to match the data from 2014-19.

Shankar et.al [4] too implemented an ARIMA (2,1,2) model but this time compared it with MLP, CNN, and LSTM models. The authors concentrated their study on Indian gold prices sourced from the same WGC dataset over ~ 5 years (January 2016 to December 2020). The LSTM-based approach yielded an RMSE of 35.87 and an MAE of 29.31, whereas the ARIMA model yielded an RMSE of 110.55 and an MAE of 97.07. Their analysis revealed that the deep learning models outperform the univariate Time Series model by a long shot. This can be explained by the inputs the researchers gave to the deep learning-based models, which were: 1) the Previous day's price of gold in the same week and 2) the Current gold price. This allowed the neural networks to much better predict the current prices of gold, as they were trained to respond to the latest prices. As such, the deep learning model would've not provided accurate results if made to predict data for the next 5 years.

Yang et.al [5] on the other hand, applied an ARIMA (3, 1, 2) model and trained their model on 6 years of data [July 1st, 2013 to June 29th, 2018] from the WGC and attempted to forecast the gold prices for the first half of July 2018. Their forecasts had a very small relative error, not Shankar et.al [4] too implemented an ARIMA (2,1,2) model but this time compared it with MLP, CNN, and LSTM models. The authors concentrated their study on Indian gold prices sourced from the same WGC dataset over ~ 5 years (January 2016 to December 2020). The LSTM-based approach yielded an RMSE of 35.87 and an MAE of 29.31, whereas the ARIMA model yielded an RMSE of 110.55 and an MAE of 97.07. Their analysis revealed that the deep learning models outperform the univariate Time Series model by a long shot. This can be explained by the inputs the researchers gave to the deep learning-based models, which were: 1) the Previous day's price of gold in the same week and 2) the Current gold price. This allowed the neural networks to much better predict the current prices of gold, as they were trained to respond to the latest prices. As such, the deep learning model would've not provided accurate results if made to predict data for the next 5 years.

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Instead of using time-series or deep learning approaches, Dr.Abhay Kumar et.al [5] used a simple linear regression-based model on the ETF Gold dataset, which had 15 years' worth of Gold ETF Rate data. The regression equation was as

follows : $\text{GoldETFPPrice} = (m1 \times 5 \text{ day moving average}) + (m2 \times 10 \text{ day moving average}) + C$

Through this equation, the authors were able to accurately predict the gold price of the next 3 days with an R2 value of 98.76%. This model, just like the previously discussed deep learning approach by Shankar et.al has a major disadvantage, in that it can only predict values of the very short future and won't work for anything above 10 10 days from the current day of prediction.

III. METHODOLOGY:

The proposed model has been built using python along with the following of its libraries: NumPy, Pandas, Matplotlib, Seaborn and SciKit Learn for machine learning. For analysis we have used correlation and the random forest model has been used for predictions.

The dataset provides gold price data over a period of ten years. First, data pre-processing was conducted in which columns such as data were dropped. The next stage was the analysis of features to determine the ones with the highest correlation and their exact relationship with the price of gold. A train-test split of 70:30 was used. In the dataset used, we have the five important parameters. SPX, which is the stock market index known as S&P 500 that tracks the top 500 companies which are publicly traded in the United States of America. The second one is USO which is the United States Oil price. Next is SLV which is the price of silver. The last one is the currency pair of EUR/USD signifying the exchange rate on that date. We trained the proposed model using these features to predict the price of gold,

We employed correlation to conduct data analysis. Correlation describes the relationship between two or more variables. A statistical technique for figuring out how one variable moves or changes in relation to another is correlation. It gives us an idea of how closely the two variables are related. Two categories of correlations exist:

Negative correlation: A relationship between two variables where one increases while the other lowers and vice versa is known as a negative correlation.
 Positive correlation: Two variables that move in tandem, or in the same direction, are said to have a positive correlation. When one variable rises while the other rises, or when one falls while the other falls, there is a positive correlation.

On calculating the correlation values of gold, we saw that the USO and the EUR/USD were slightly negatively correlated whereas the others were positively correlated.

The features and the target values were then split. The target is the gold price and the features are all the stock prices. A variable X is used to store all the features by dropping the date and gold columns, and Y is used to store the gold price.

Further, we had perform the split into the training and testing data. For this, we created four variables: X_train, X_test, Y_train, Y_test. 70% of the values went in X_train and 30% in X_test. The same split was followed for Y_train and Y_test.

The next part of the process is training the core model for which we have chosen the Random Forest Regressor algorithm. The algorithm is an ensemble of n-decision tree models (n = 100 for our model) which each give an output. The output is then collated together in a process known as

“Majority Voting” using statistical techniques such as mean, median or mode. The idea behind the algorithm is that instead of concentrating the dependence on the model’s prediction a single decision tree, we leverage ensemble learning and aggregate numerous trees to predict the final result.

We have used Random Forest algorithm because it is a popular choice for gold price prediction using machine learning for several reasons:

1) **Robustness:** The robustness of Random Forest against overfitting is well-known, and this is important when working with financial data such as gold prices. When a model learns noise from training data instead of the underlying pattern, it is said to be overfitting and performs poorly on untrained data.

2) **Managing Non-linearity:** The price of gold frequently demonstrates intricate non-linear correlations with a range of variables, including market emotions, geopolitical events, and economic indicators. Because Random Forest is an ensemble model, it can successfully capture these non-linear interactions, which makes it a good choice for modeling the complex dynamics of gold prices.

3) **Feature Importance:** Accurate forecast requires an understanding of the factors impacting gold prices. A metric of feature importance is provided by Random Forest, highlighting the factors that have the greatest influence on the prediction. This can help with risk management and decision-making by providing insightful information about the factors influencing changes in the price of gold.

4) **Managing High-Dimensional Data:** Financial datasets frequently have a lot of variables, some of which might be superfluous or unnecessary. Modeling is made easier using Random Forest since it can handle high-dimensional data well without the requirement for feature selection or dimensionality reduction methods. The random forest regression model's benefits include its strong generalization and prediction capabilities, as well as its ability to successfully prevent overfitting issues. The model adds randomization, and its prediction accuracy and stability can be further enhanced by modifying the regression trees' hyperparameters.[15]

5) **Simple to build and Tune:** Compared to other sophisticated algorithms like neural networks, Random Forest is comparatively simple to build and requires less hyperparameter adjustment. Because of this, practitioners with limited computational resources and experience in machine learning can still use it.

Using random forests helps us with reducing overfitting and works great with categorical or continuous values. We didn’t have to scale the data either unlike other ML algorithms since it uses a rule-based approach. It's noteworthy to notice that both classifiers work well with smaller training data (i.e., 45-75% training set version), despite the fact that ML algorithms often perform better when greater training data is used[9].

Also, since we are predicting gold price, some unexpected turns in the economy may cause the price of gold to rise or fall in a very short span of time. Thankfully, random forests are quite stable and therefore do not get heavily impacted by certain outlier events

To use the algorithm we are using the random forest function present in the sklearn library by calling the regressor.fit() through which our model will be trained.

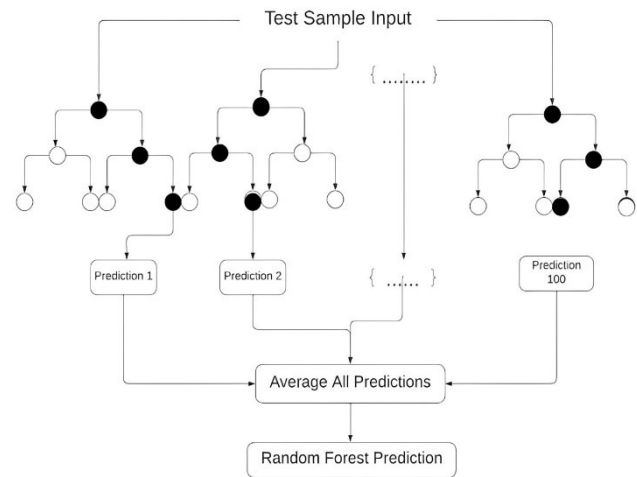


Fig. 1 Test Sample

As per figure 1 Random Forest prediction is obtained by averaging multiple predictions generated by the model for each observation in the test sample data. This approach helps enhance the stability and reliability of the predictions, contributing to a more accurate estimation of gold prices based on the input features.

We then evaluate the model based on the test data. To do that, we call regressor.predict() function on the test data and generate the predicted gold prices. For comparing the predicted values and the actual values we have calculated the R-squared score also known as the co-efficient of determination[16]. To determine how the value of the dependent variable varies when one of the independent factors changes while the other variables remain stable, regression analysis is utilized.[13] The value tells us how well our model accounts for the variance in the test data. If our model adjusts well to the variance in the data points aka. when the data points are closer to the regression line created by our model, the r-squared value is closer to 100%. We managed to achieve a r-squared value of 98.94%.

We've also plotted an actual price vs expected price graph, which shows that the actual and predicted prices are fairly similar to each other.

Methods to improve performance & Accuracy:

To improve the performance of your Random Forest model for gold price prediction, we considered the following methods:

1) **Feature engineering:** we looked at adding new features or changing current features to perhaps capture more data related to changes in the price of gold.

2) we utilized cross-validation methods, such as k-fold cross-validation, to lessen overfitting and more accurately estimate the model's performance on unknown data. This entails dividing the data into several subsets, using each subset to train the model, and then assessing the model's performance on the remaining subset.

We paid attention to data preprocessing steps such as handling missing values, scaling numerical features, and encoding categorical variables. Ensured that the data is clean and properly formatted before training the model.

We identified and removed outliers from the dataset that may adversely affect model performance. Outliers can distort patterns and relationships in the data, leading to suboptimal predictions.

Dataset

	A	B	C	D	E	F
1	Date	SPX	GLD	USO	SLV	EUR/USD
2	1/2/2008	1447.160034	84.860001	78.470001	15.18	1.471692
3	1/3/2008	1447.160034	85.57	78.370003	15.285	1.474491
4	1/4/2008	1411.630005	85.129997	77.309998	15.167	1.475492
5	1/7/2008	1416.180054	84.769997	75.5	15.053	1.468299
6	1/8/2008	1390.189941	86.779999	76.059998	15.59	1.557099
7	1/9/2008	1409.130005	86.550003	75.25	15.52	1.466405
8	1/10/2008	1420.329956	88.25	74.019997	16.061001	1.4801
9	1/11/2008	1401.02002	88.580002	73.089996	16.077	1.479006
10	1/14/2008	1416.25	89.540001	74.25	16.280001	1.4869
11	1/15/2008	1380.949951	87.989998	72.779999	15.834	1.48021
12	1/16/2008	1373.199951	86.699997	71.849998	15.654	1.466405
13	1/17/2008	1333.25	86.5	71.029999	15.717	1.464
14	1/18/2008	1325.189941	87.419998	71.540001	16.030001	1.461796
15	1/22/2008	1310.5	88.169998	70.550003	15.902	1.464794

Fig. 2 Dataset

As per figure 2 Dataset Name: Gold Price Dataset

Total Records: The dataset contains records from January 2, 2008, to May 16, 2018.

Source: The dataset is sourced from Kaggle. It is financial data related to stock market indices (e.g., SPX), commodity prices (e.g., GLD, USO, SLV), and currency exchange rates (e.g., EUR/USD). This type of data is commonly sourced from financial databases, economic research institutions, or public datasets provided by financial organizations.

Model Evaluation:

- 1) We used a 70:30 ratio to split the dataset into training and testing sets.
- 2) Using the features (SPX, USO, SLV, EUR/USD) in the training data, we trained the Random Forest regression model to predict the target variable (gold price).
- 3) We assessed the model's performance using the R-squared score, commonly referred to as the coefficient of determination. The percentage of variance in the target variable (gold price) that can be predicted from the independent variables (features) is measured by the R-squared score. Better predictive performance is indicated by a higher R-squared score; a maximum value of 1 denotes a perfect match.
- 4) To visually compare the projected and actual gold values in the test dataset, we created a "Actual Price vs. Predicted Price" graph. This graph aids in evaluating the degree to which the true values and expected values coincide.

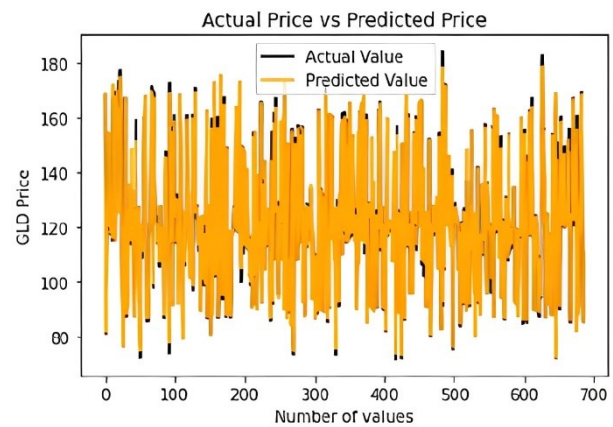


Fig. 3 Actual Price vs Predicted Price
Number of Values

We achieved the figure 3 output in the following way :

- We had the actual gold prices available in your test dataset, which serve as the ground truth values for comparison.
- To visually compare the predicted gold prices with the actual prices, we created a scatter plot where:
- The x-axis represents the actual gold prices from the test dataset.
- The y-axis represents the predicted gold prices generated by your Random Forest model.
- Each data point in the scatter plot corresponds to a specific date or time period in our test dataset. We plotted the actual gold prices against the predicted prices for each corresponding date or time period.
- By examining the scatter plot, we assessed how closely the predicted gold prices align with the actual prices. Ideally, the data points should fall close to a diagonal line (i.e., the line of perfect prediction), indicating a strong correspondence between the predicted and actual values.

IV. RESULTS AND DISCUSSION

Blue represents train data and pink represents test data in Figure 3. With 98.9% accuracy, it shows the most promising results. The simple linear regression model as shown in Figure 4, on the other hand, has a 75.3 percent accuracy. The accuracy of the Support Vector Regression model is 94.7 percent. Through these results we can conclude that random forest model provides the best results on the given data.

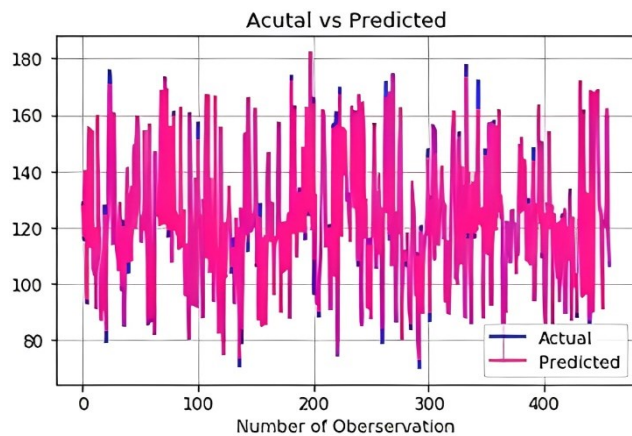


Fig. 4 Actual Price vs Predicted Number of Observation

As per figure 4 each data point on the plot represents a pair of actual gold prices (from the test dataset) and predicted gold prices (from the model) for a particular observation. It indicates different way of organizing or categorizing the data points for visualization purposes.

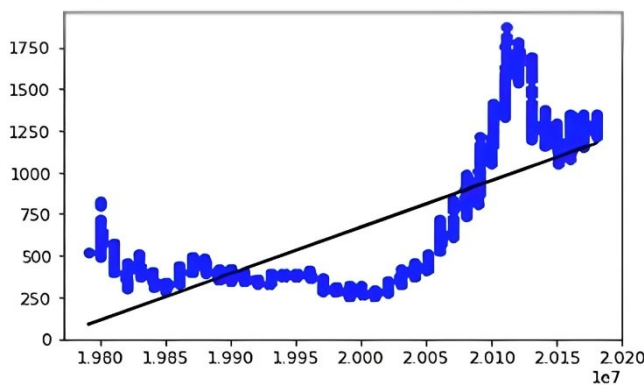


Fig. 5 Test Data

As per figure 5 visual representation of the test dataset, which contains a subset of the overall data used in our analysis. The test dataset is typically separate from the training dataset and is used to evaluate the performance of the machine learning model.

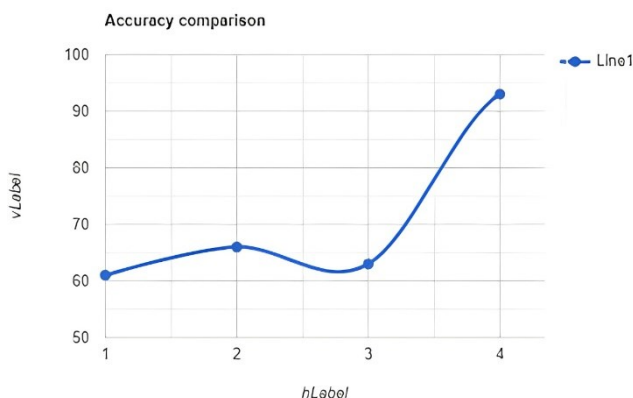


Fig. 5 Accuracy Comparison

	Author	Method	Accuracy
1	V. K. F. B. Rebecca Davis	Autoregressive Moving Average (ARMA) mode	66.66%

2	Samar P. Patil, V. M. Vrasava and G. M. Pooddar	Future of silver, cu future, oil future, Euro-USD's, SIP/TAX combined, use algorithmic like SVMs Logistics regression for forecast.	61.91%
3	M. S. Kuumar Chanuar	Extreme learning machines(ELM) algorithm, a variation of ANN	93.82%
4	Our Model	Random Forest Algorithm(XL)	98.9%

Table 1: Comparison with Earlier Models

V. CONCLUSION:

The purpose of this study was to investigate the relationship between the price of gold and a few key determinants, including the stock market, the price of crude oil, the rupee to dollar exchange rate, inflation, and interest rates. Three machine learning algorithms—linear regression, random forest regression, and gradient boosting regression—were used to analyze these data. A more potential option for performing forecasting and prediction analysis is the support vector machine. It is discovered that gradient boosting regression outperforms random forest regression in terms of prediction accuracy over the course of the two periods when analyzed separately.

The investigation comes to the conclusion that machine learning algorithms are highly helpful in this type of analysis, although their accuracy depends on the data's properties. A confusion matrix is used to assess each model's correctness once it has been trained and tested. For a better understanding of how these strategies operate, additional research using these data and various techniques may be undertaken.

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